Lab Assignment 8

In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

In [3]:
```
df = pd.read_csv("titanic.csv")
```

In [4]:
```
df
```

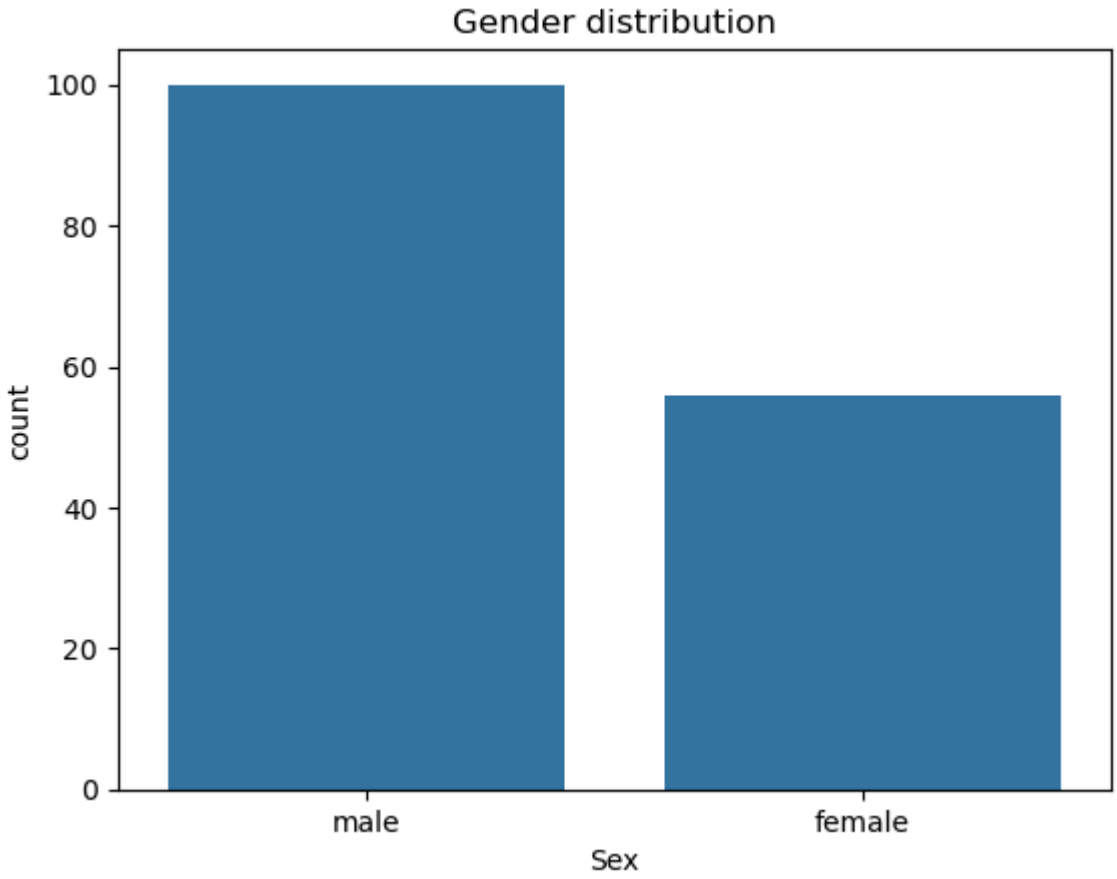
Out[4]:

	PassengerId	Survived	Pclass	Lname	Name	Sex	Age	SibSp	Parch	
0	1	0	3	Braund	Mr. Owen Harris	male	22.0	1	0	
1	2	1	1	Cumings	Mrs. John Bradley (Florence Briggs Thayer)	female	38.0	1	0	P
2	3	1	3	Heikkinen	Miss. Laina	female	26.0	0	0	ST 3
3	4	1	1	Futrelle	Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	
4	5	0	3	Allen	Mr. William Henry	male	35.0	0	0	
...	...	...	...	...	...	...	...	...	...	
151	152	1	1	Pears	Mrs. Thomas (Edith Wearne)	female	22.0	1	0	
152	153	0	3	Meo	Mr. Alfonzo	male	55.5	0	0	
153	154	0	3	van Billiard	Mr. Austin Blyler	male	40.5	0	2	A
154	155	0	3	Olsen	Mr. Ole Martin	male	NaN	0	0	
155	156	0	1	Williams	Mr. Charles Duane	male	51.0	0	1	P

156 rows × 13 columns

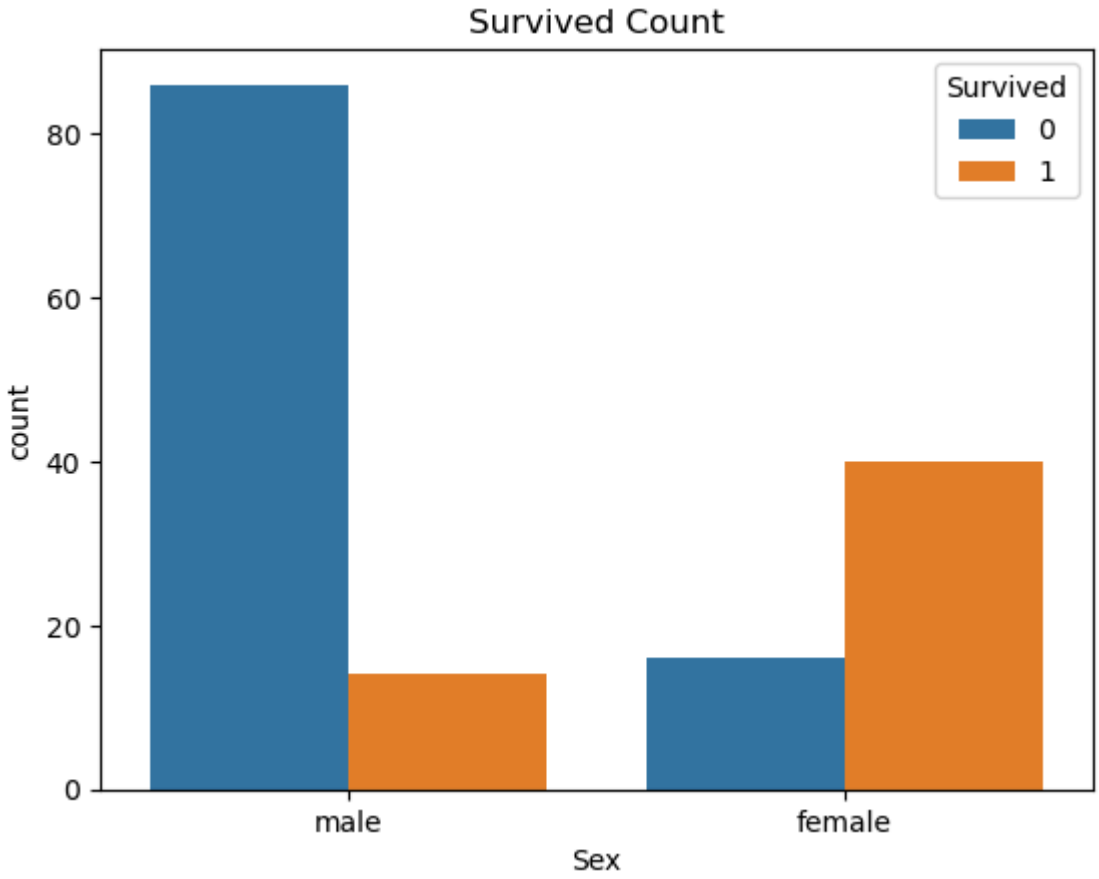
In [5]:

```
sns.countplot(x = 'Sex',data = df)
plt.title("Gender distribution")
plt.show()
```



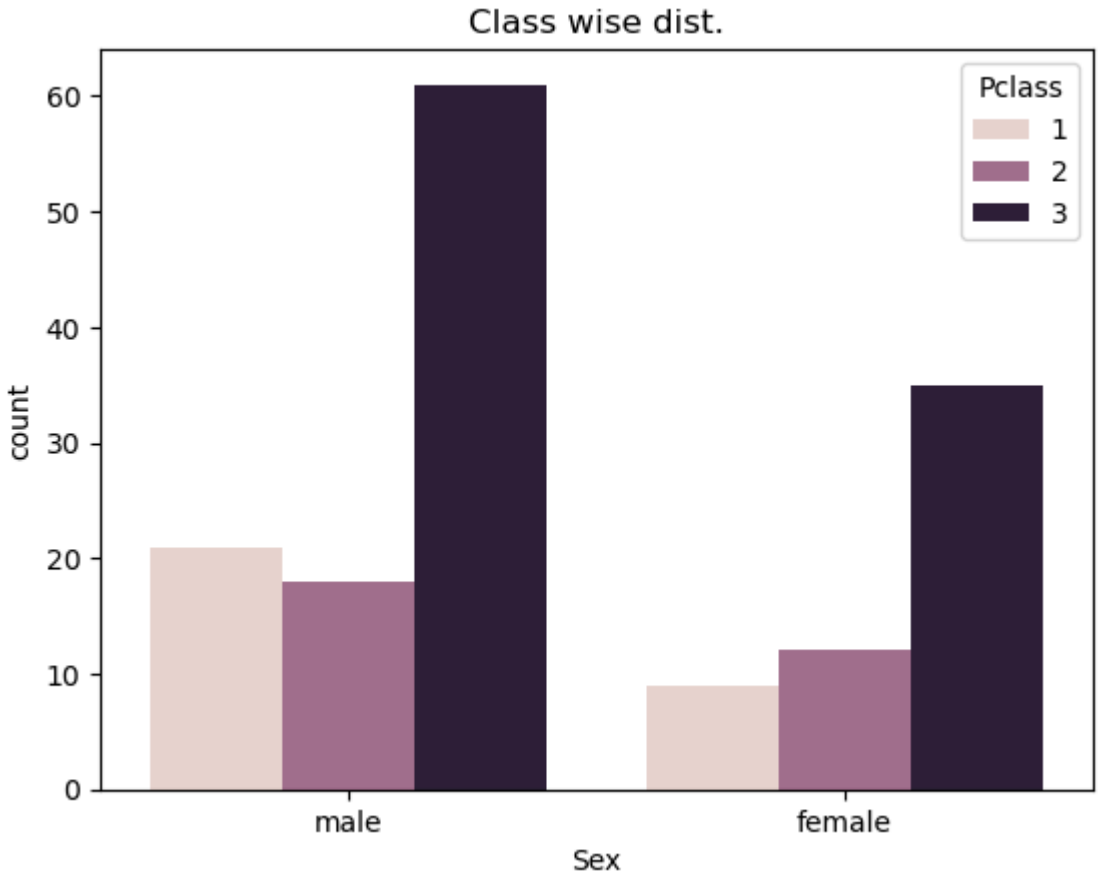
Observation : Male passengers are more than female

In [6]:
```
sns.countplot(x = 'Sex',hue = 'Survived',data = df)
plt.title("Survived Count")
plt.show()
```



Observation : Female Survivors are more than male survivors

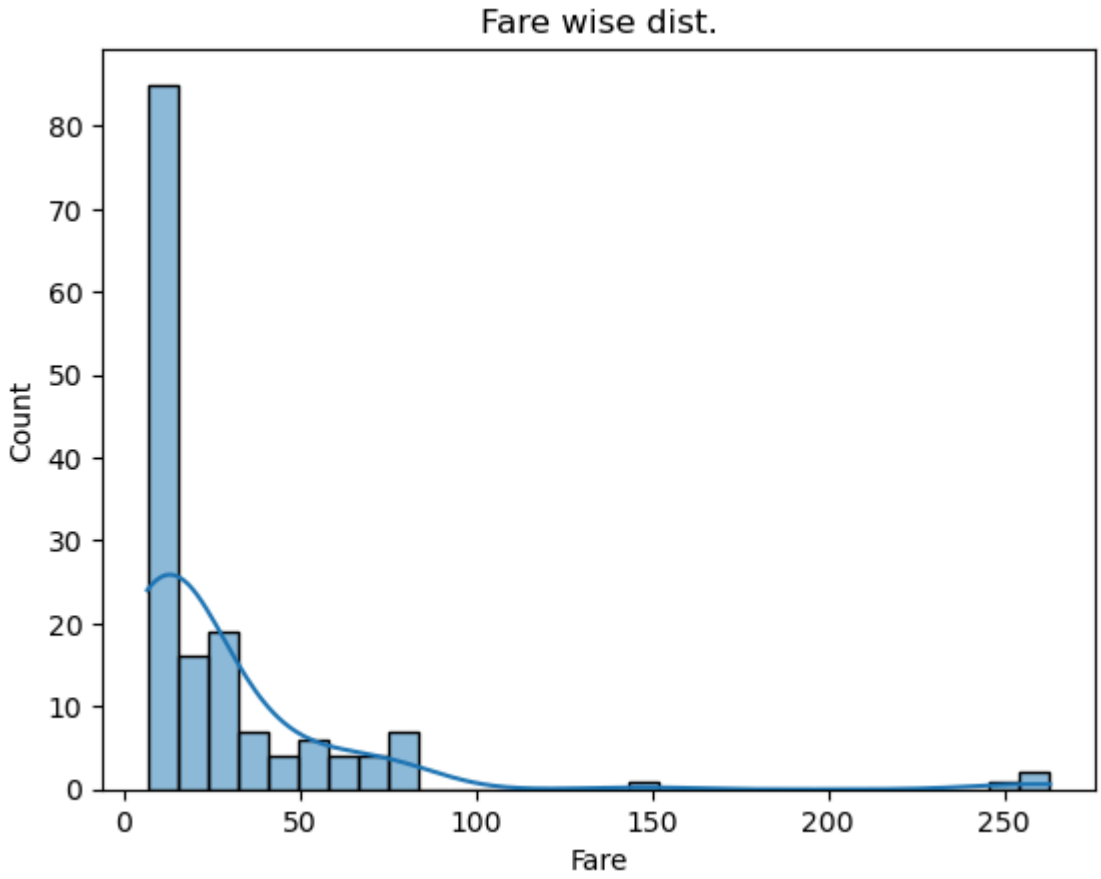
In [7]:
```
sns.countplot(x = 'Sex',hue = 'Pclass',data = df)
plt.title("Class wise dist.")
plt.show()
```



Observation 1: 3rd Class passenegers are more than other classes

Observation 2: 1st Class passenegers are more than 2nd class passengers

In [8]:
```
sns.histplot(df['Fare'],bins = 30,kde = True)
plt.xlabel("Fare")
plt.title("Fare wise dist.")
plt.show()
```



Observation 1: The graph is right skewed

Observation 2: Majority of values are concentrated in the lower fare range (0–50)

Observation 3 : The data is not normally distributed